HILAL BEYZA TASDEMIR

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Software engineer who takes full-stack and ML systems from prototype to production, focused on computer vision and security.

EDUCATION

Purdue University, Elmore Family School of Electrical and Computer Engineering **West Lafayette, IN**
Bachelor of Science in Computer Engineering *Aug 2022 – May 2026*

- **Achievements:** Certificate of Entrepreneurship, Concentration in Software Engineering, Minor in Mathematics
- **Relevant Coursework:** Software Engineering, Artificial Intelligence, Computer Security, Data Structures

PROFESSIONAL EXPERIENCE

Navigation Lead, Stride - Accessibility App for Visually Impaired Users *Aug 2025 – May 2026*

- Enabled end-to-end Dijkstra routing across an entire test building by building an **automated, graph-based data ingestion pipeline** that turned hand-measured floor plans into a routable graph of 76 nodes, 76 edges, and 79 landmarks across 2 floors, with coordinate transforms, bearing calculations, and data-validation scripts.
- Took the **AWS serverless backend** from prototype to production (API Gateway, WebSocket, Lambda, RDS, IAM, CloudWatch) powering real-time indoor navigation, with automated CDK deployment workflows for repeatable releases.
- Integrated ADA/WCAG-compliant collision detection built to a **100 ms response-time requirement**, validated through 2 rounds of user acceptance testing with blindfolded participants simulating real accessibility conditions.
- Fixed a routing bug found in first-round testing, where users were sent past their destination and back, by moving door nodes from wall-aligned edges to hallway centerlines and adding left/right metadata before round 2.

Systems Engineering Intern, ZF Group *May 2024 – Aug 2024*

- Automated truck positioning for binding operations, previously done by manual alignment, by building an **OpenCV computer vision** pipeline that used Gaussian filtering, Canny edge detection, and Hough Line Transform to detect alignment features.
- Engineered **hardware-in-the-loop** testbench using IQAN to simulate truck steering dynamics across 2 driver-assist modes, eliminating the need for on-vehicle testing during control optimization.

Undergraduate Teaching Assistant, Purdue University - ECE 36200 & ECE 57000 *Aug 2025 – May 2026*

- Selected as an undergraduate TA for a **graduate-level AI course** (ECE 57000), supporting instruction and grading for ML topics.
- Assisted programming labs for 200+ students in ECE 36200, covering embedded systems fundamentals in **Assembly & C**.

PROJECTS

Semiconductor Defect Detection Using Vision Transformers

- Fine-tuned a pre-trained **Vision Transformer** (ViT-Base) for 9-class wafer-map defect classification from only 631 training images, an estimated **90% less labeled data** than state-of-the-art CNNs, reaching **87.7% accuracy** and **0.876 macro F1**. A two-stage fine-tuning strategy with Focal Loss raised accuracy **3.9 points** over a standard ViT baseline.

BoilerFace, ML Model Registry Platform

- Co-built an **LLM-assisted** model evaluation system that scored open-source ML models on quality and compliance metrics (such as data quality, code quality), integrating **Hugging Face** and **GitHub REST APIs** to automate a previously manual vetting process, backed by **76 pytest tests at 82% coverage**.
- Extended it into a **FastAPI** model registry, adding reproducibility and tree-score metrics and a WCAG 2.1 AA web interface verified with automated Selenium accessibility tests.

Calculator0, AI Financial Exit Planner for Abuse Survivors - Hackathon

- Built a disguised app (appearing as a calculator) for financial abuse survivors using **React Native/TypeScript**, **Flask**, and **Firebase**, integrating **Gemini AI** to analyze their bank transactions and generate personalized financial exit plans.
- Encrypted survivors' financial data with **AES-256** as a first step toward **zero-knowledge** storage, and added a **decoy PIN** that opens fake data if someone forces the user to unlock the app.

LEADERSHIP

Purdue Muslim Student Association *May 2024 – May 2026*
President, *May 2025 – May 2026* / ***Treasurer,*** *May 2024 – May 2025*

- Led a 14-member board for a 200+ member organization, growing event attendance by **60%** while expanding the organizational scope into professional networking, interfaith, and civic engagement events.
- Secured the SOGA grant 5 times, raising **$15,000+** in external funding across two years to sustain 30+ events per semester.
- Managed **$20,000+** in cumulative organizational finances, overseeing budgeting, allocation, and reporting for all MSA initiatives.
- Received the **Dean Betty Nelson Student Organization Service Award 2026**, awarded to 1 organization university-wide each year.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C, C++, SQL
Frameworks & Libraries: React Native, FastAPI, Flask, Node.js, OpenCV, OpenGL, Three.js, Hugging Face, SQLite, Pytest, Selenium
Cloud & Tools: AWS (CDK, RDS, Lambda, EC2), GCP, Firebase, Docker, Git, GitHub Actions, CI/CD, VS Code, Google Colab, Jira